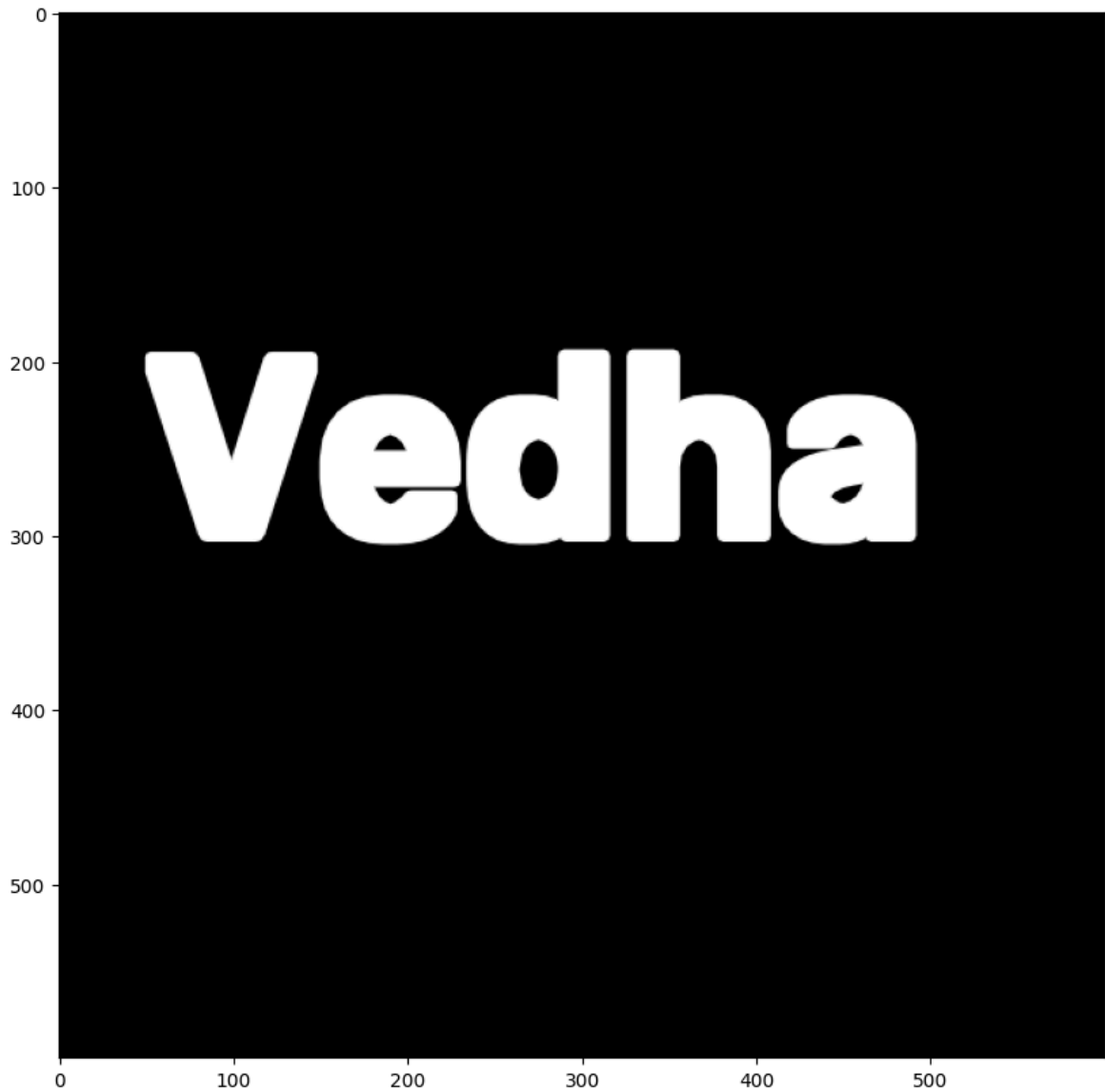




Ex. No: 10

Record-IMPLEMENTATION OF OPENING AND CLOSING

Name: Vedha M Reg. No: 212225230292 Slot: 26OD1199

OPENING

```
In [8]: img = load_img()
```

```
In [9]: white_noise = np.random.randint(low=0,high=2,size=(600,600))
```

```
In [10]: white_noise
```

```
Out[10]: array([[1, 1, 0, ..., 0, 0, 0],
               [1, 0, 1, ..., 1, 0, 0],
               [0, 1, 0, ..., 1, 1, 0],
               ...,
               [0, 1, 0, ..., 0, 0, 1],
               [1, 1, 0, ..., 0, 1, 1],
               [0, 0, 0, ..., 1, 1, 0]], shape=(600, 600), dtype=int32)
```

```
In [11]: img.max()
```

```
Out[11]: np.uint8(255)
```

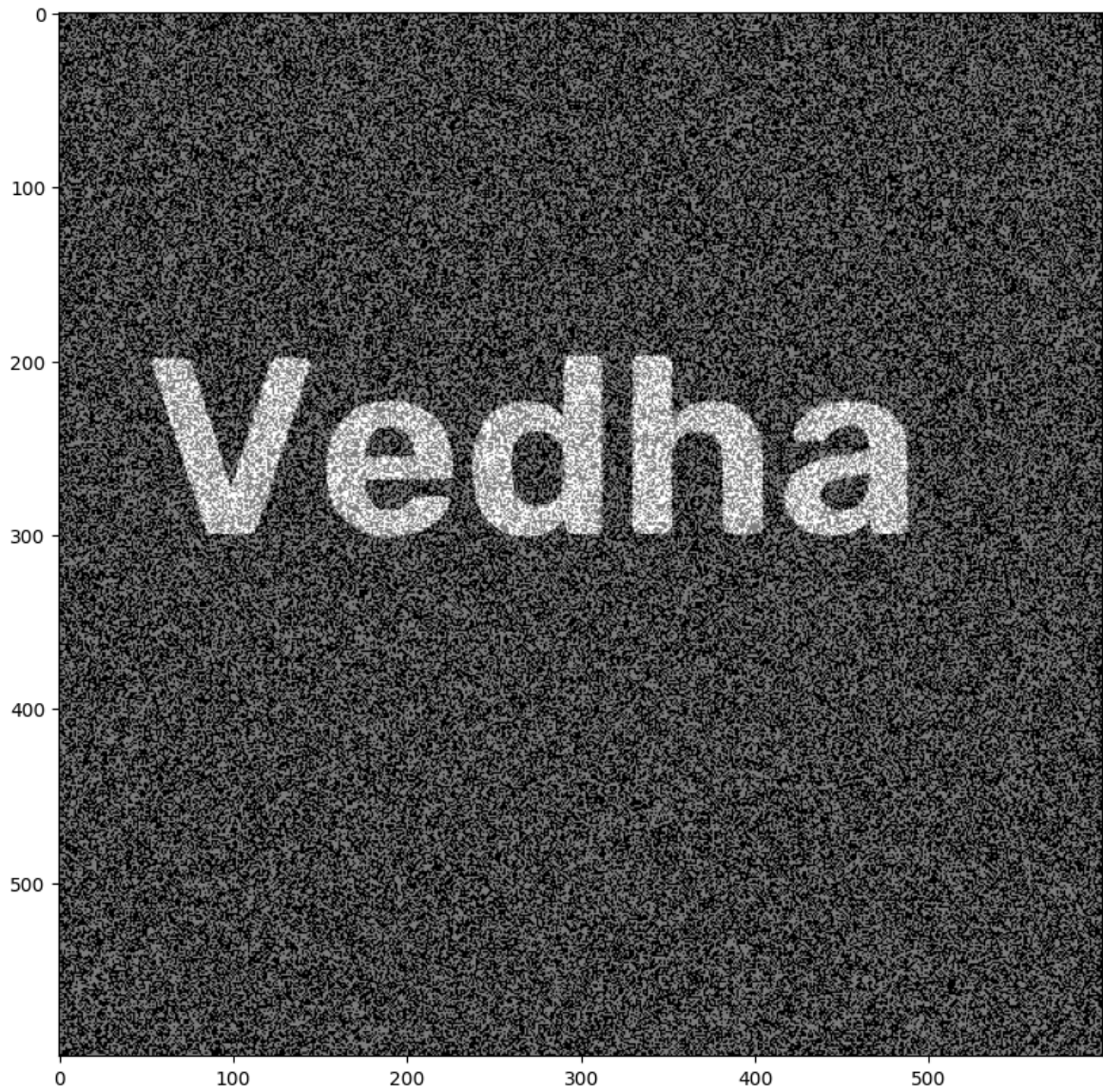
```
In [12]: white_noise = white_noise*255
```

```
In [13]: white_noise
```

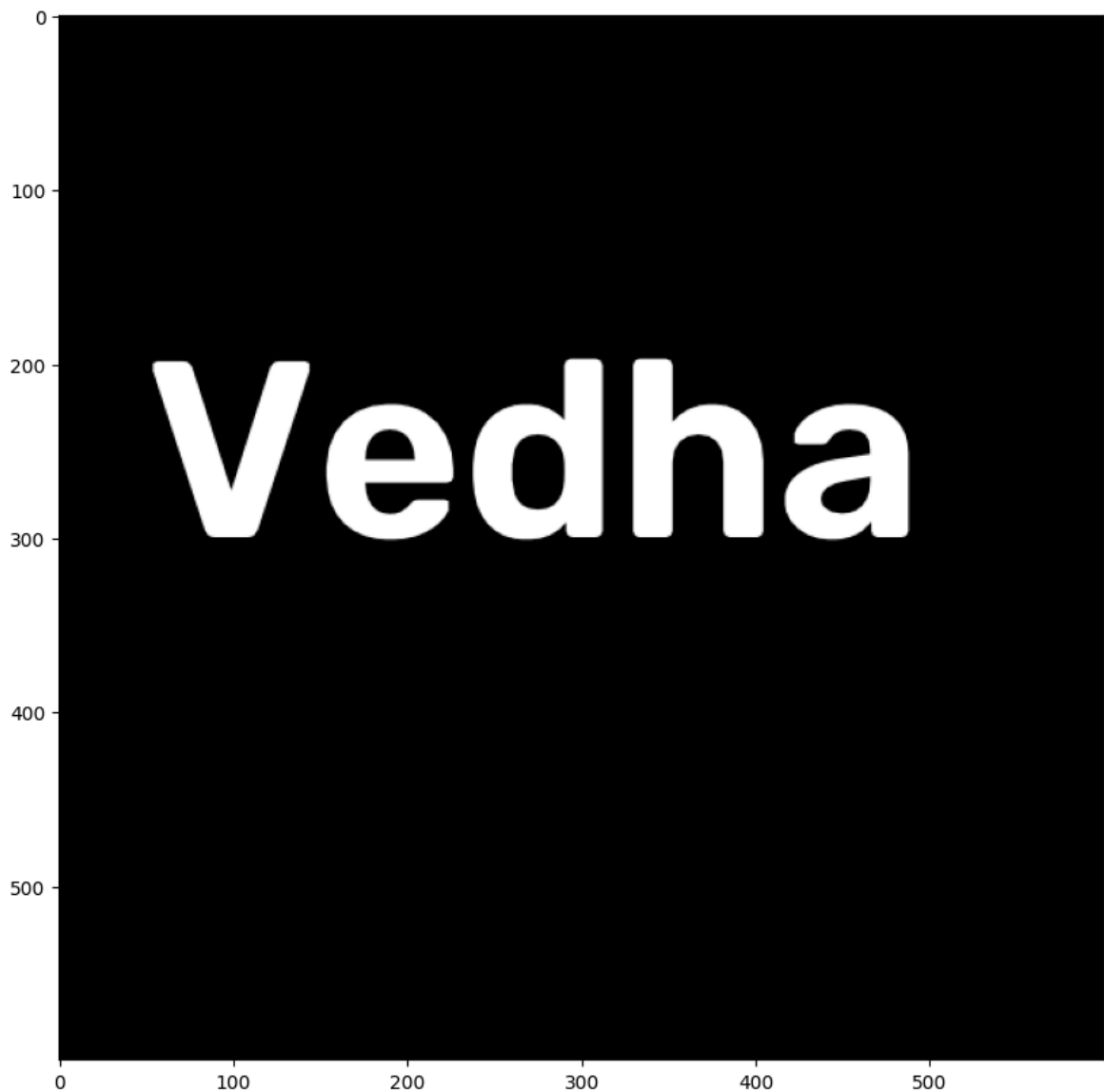
```
Out[13]: array([[255, 255,  0, ...,  0,  0,  0],
               [255,  0, 255, ..., 255,  0,  0],
               [ 0, 255,  0, ..., 255, 255,  0],
               ...,
               [ 0, 255,  0, ...,  0,  0, 255],
               [255, 255,  0, ...,  0, 255, 255],
               [ 0,  0,  0, ..., 255, 255,  0]], shape=(600, 600), dtype=int32)
```

```
In [14]: noise_img = white_noise+img
```

```
In [15]: display_img(noise_img)
```



```
In [16]: opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)
display_img(opening)
```



CLOSING

```
In [17]: img = load_img()
```

```
In [18]: black_noise = np.random.randint(low=0,high=2,size=(600,600))
```

```
In [19]: black_noise
```

```
Out[19]: array([[1, 0, 1, ..., 0, 1, 0],
                [0, 1, 1, ..., 0, 1, 1],
                [0, 0, 0, ..., 1, 1, 1],
                ...,
                [1, 1, 1, ..., 0, 0, 1],
                [1, 0, 1, ..., 0, 0, 0],
                [1, 1, 1, ..., 0, 0, 0]], shape=(600, 600), dtype=int32)
```

```
In [20]: black_noise= black_noise * -255
```

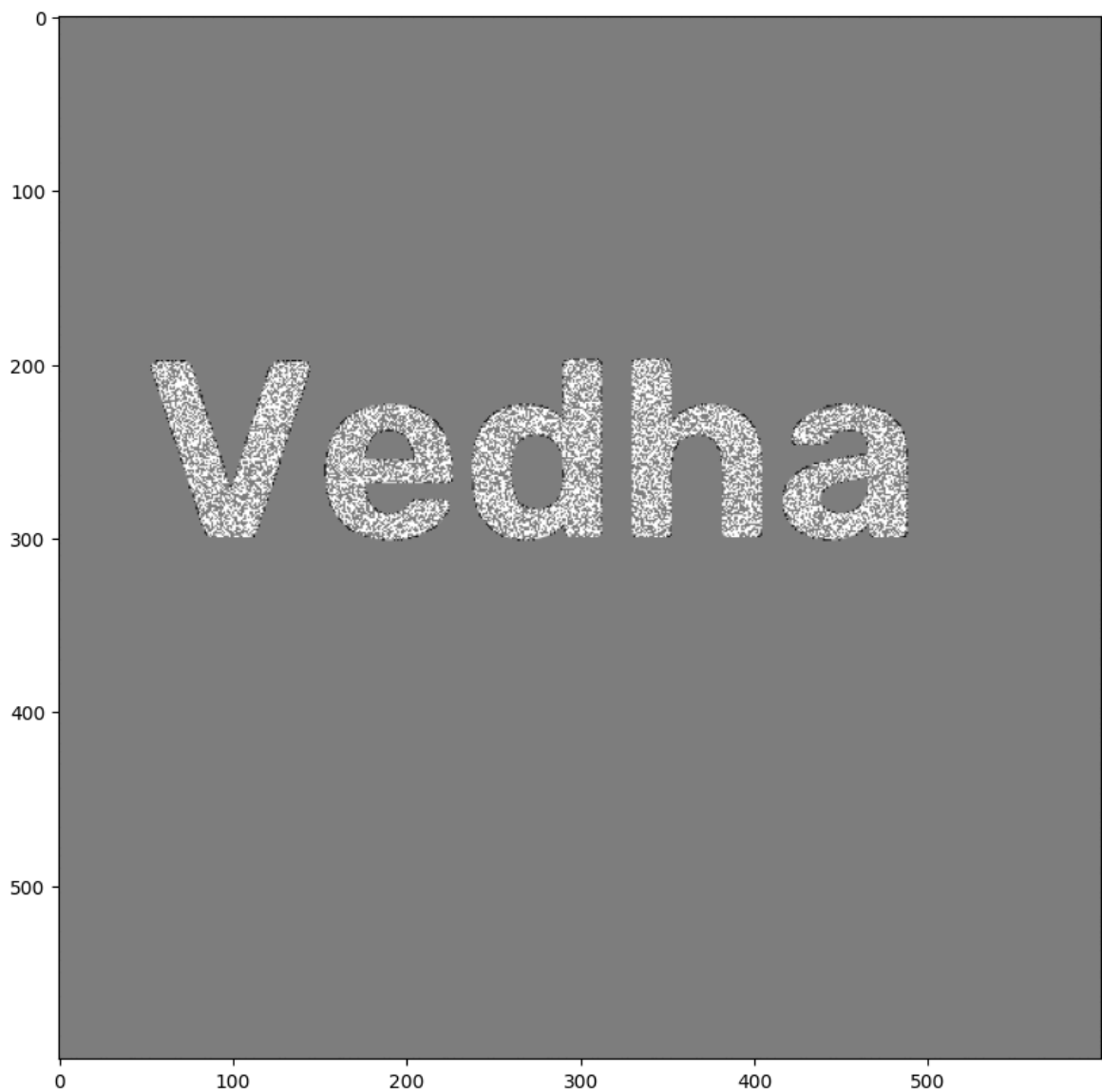
```
In [22]: black_noise_img = img + black_noise
```

```
In [23]: black_noise_img
```

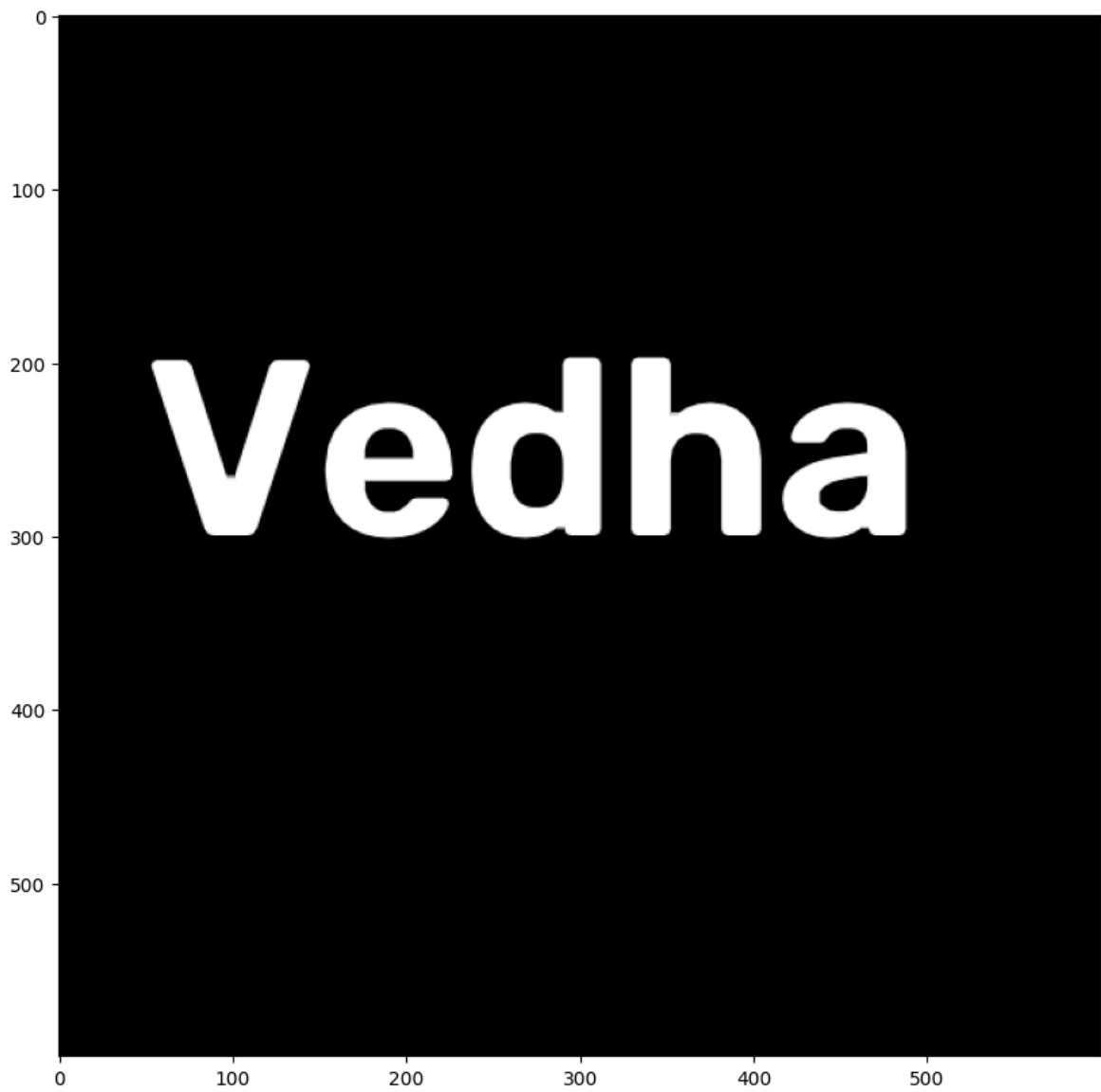
```
Out[23]: array([[ -255,    0, -255, ...,    0, -255,    0],
               [    0, -255, -255, ...,    0, -255, -255],
               [    0,    0,    0, ..., -255, -255, -255],
               ...,
               [-255, -255, -255, ...,    0,    0, -255],
               [-255,    0, -255, ...,    0,    0,    0],
               [-255, -255, -255, ...,    0,    0,    0]],
              shape=(600, 600), dtype=int32)
```

```
In [24]: black_noise_img[black_noise_img== -255] = 0
```

```
In [25]: display_img(black_noise_img)
```



```
In [26]: closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)
         display_img(closing)
```



In []: